

## **1. Introduction: From Free Will to Sense of Agency**

The problem of free will has run through the history of Western thought like a red thread: are we, as individuals, the authors of our actions, or are they determined by external causes? Psychology does not aim to solve this philosophical problem, but it investigates a related concept that is far more empirically accessible: the sense of agency (SoA). This is defined as the subjective experience of being the initiator and controller of one's own actions and of their effects on the environment (Haggard & Tsakiris, 2009). Three fundamental aspects underpin SoA: intention (the desired state that one pursues through action), action (awareness of one's own actions), and effect (perception of the consequences of action) (Wen & Imamizu, 2022). These three components are directly linked to the motor, perceptual, and sensorimotor systems, each playing a major role in generating SoA.

It is important to emphasize from the outset that SoA does not refer to a global attitude towards life, but to specific experiences tied to isolated events or particular action sequences. Most experiments in this field focus on such micro-events. In the laboratory, SoA is typically studied by introducing a causal relationship between the participant's action and a sensory effect, in which the experimenter distorts either the temporal or the spatial relationship between the two (Farrer et al., 2003; Metcalfe & Greene, 2007; Sato & Yasuda, 2005; Moore et al., 2009). The participant must then judge whether he or she caused the observed effect. This type of design allows for fine manipulation of SoA and its systematic measurement.

SoA is not limited to one's own body. For example, when a person uses buttons to control an avatar in a video game, although the avatar's detailed movements are generated by the computer, that person feels a sense of agency over the avatar because its movements match the buttons pressed. If, however, a computer error occurs, the person may temporarily lose SoA due to the sudden mismatch between the avatar's movement and the buttons pressed (Wen & Imamizu, 2022). This example illustrates the fundamental principle of SoA: the match between action and effect.

## **2. Distinguishing SoA from the Sense of Ownership**

SoA must be distinguished from the sense of ownership (SoO), which refers to the awareness that one's own body and its parts belong to oneself (Braun et al., 2018; see also Gallagher, 2000; Tsakiris et al., 2007). Although the two are closely related, they can be dissociated: I can feel that my leg belongs to me (ownership) without feeling that I initiated its movement (absence of agency). Such dissociations occur both in everyday life — for example, in the case of tics or involuntary movements — and in clinical conditions such as Tourette syndrome, anarchic hand syndrome, or somatoparaphrenia (Tsakiris, 2010). Furthermore, in the rubber hand

illusion, participants can feel that the fake hand belongs to them (SoO) without feeling that they control it (SoA).

The importance of SoA lies in the fact that it allows us to monitor our own actions and to distinguish them from external causes. When we act, the brain constantly compares the predicted sensory consequences with the actual feedback; a mismatch generates an error signal that enables us to adjust our behaviour and reduces the sense of agency (Engbert et al., 2008; Frith, Blakemore & Wolpert, 2000). In addition, SoA plays a central role in social interaction, allowing us to discern whether an outcome was produced by us or by our partner.

### **3. Theoretical Models of SoA**

Four major theoretical models have been proposed to explain the generation of SoA. These models are not mutually exclusive, but complementary: together they provide a comprehensive framework that spans from simple sensorimotor mechanisms to complex cognitive inferences.

#### **3.1 The Comparator Model**

The comparator model is the classical account of SoA. According to it, the motor control system uses five types of representations (Carruthers, 2010; Frith, Blakemore & Wolpert, 2000): (a) the goal state — the desired final position of the body, also called motor intention; (b) the motor commands, of which two copies are produced through the inverse model; the copy that remains central is called the corollary discharge or efference copy; (c) the predicted sensory consequences, generated by the forward model based on the efference copy; (d) the actual sensory consequences of the movement (reafference); (e) an estimation of body position, resulting from the continuous combination of predicted and actual information. SoA arises when the comparison between (c) and (d) indicates a match; when the two do not coincide, the event is perceived as external, and SoA is reduced or absent.

However, the comparator model has significant limitations. It applies mainly to simple motor actions, with immediate and predictable sensory consequences, and does not explain SoA for complex actions, with delayed effects or multiple possible consequences. Moreover, it can only explain the implicit, low-level component of agency (FoA), not the explicit judgment (JoA), which involves conceptual inferences and social context. In addition, the model requires case-by-case adjustments to explain problematic data: in some conditions, non-visual actual sensory consequences do not appear to play a role in generating SoA, while in others, a mismatch in the auditory modality is sufficient to reduce SoA, even if visual feedback is congruent (Carruthers, 2010). This observation suggests that different sensory modalities have different weights depending on context — which led to the development of the multifactorial weighting model of Synofzik and colleagues (2008b).

### **3.2 Wegner's Retrospective Theory**

The second major theoretical model — and the one that complements the comparator model — is the retrospective (or postdictive) theory proposed by Wegner (2002). Unlike the comparator model, which explains SoA through the comparison of prediction with actual sensory feedback, the retrospective theory holds that SoA is not based on prediction, but on a post-hoc causal inference: agency is experienced when the brain infers that its own thoughts caused the action, based on contextual cues available after the outcome.

Wegner (2002) proposed that this causal inference relies on three principles that must be jointly satisfied for SoA to occur. The first is priority: the thought or intention must temporally precede the action. If the intention appears after the action, the causal link cannot be established. The second is exclusivity: there must be no other plausible causes of the action. If an external agent could have produced the action, self-attribution is reduced. The third is consistency: the observed outcome must match the initial intention. If the outcome differs from what was intended, the causal inference is weakened (Wegner, 2002; see also Wen & Imamizu, 2022). This logic mirrors causal inference in science: the cause must precede the effect, must be the only plausible cause, and must produce a consistent effect when repeated.

These three principles were demonstrated experimentally in Wegner's classic studies. In the "helping hands" experiment (Wegner, Sparrow & Winerman, 2004), participants stood with their arms at their sides while a confederate moved their arms underneath, so that the confederate's movements appeared to be the participant's. When participants heard instructions describing the confederate's movements, they reported stronger SoA over those movements — even though they did not execute them. In the "I-spy" experiment (Wegner & Wheatley, 1999), participants believed they were jointly controlling a cursor on a screen with a confederate; when the confederate forced the cursor to stop on an object, and the object's name had been mentioned 5 or 1 second earlier, participants reported stronger SoA over the stop — even though they had not controlled it. These studies demonstrate that SoA can be induced by contextual cues (priority, consistency), even in the absence of a real action.

However, the retrospective theory also has limitations. It does not explain how SoA arises for simple, automatised actions, in which there is no explicit conscious intention (e.g., involuntary postural adjustments). Nor does it explain why SoA is reduced under conditions in which sensorimotor prediction fails but the context is favorable (e.g., in the case of passive movements in Dubynin & Shishkin, 2017).

### **3.3 The Bayesian Integration Model**

Both the comparator model and Wegner's retrospective theory partially explain SoA, but neither succeeds in integrating all experimental data. This mutual limitation led to the development of a

third model — the cue integration model or Bayesian model — which proposes that SoA does not depend on a single type of cue, but on the weighted integration of multiple sources of information, both prospective and retrospective (Moore & Haggard, 2010; Synofzik, Vosgerau & Newen, 2008b).

According to this model, the brain computes the probability that an event is caused by the self ( $p(\text{self}|\text{outcome})$ ) based on a Bayesian rule: the probability is proportional to the product of the probability that the self produced that outcome ( $p(\text{outcome}|\text{self})$ ) and the prior probability that the self is the agent ( $p(\text{self})$ ), divided by the general probability of the outcome ( $p(\text{outcome})$ ) (Moore & Haggard, 2010; Wen & Imamizu, 2022). In other words, SoA is not a binary process (present or absent), but a probabilistic continuum that depends on how likely it is that one's action caused the observed effect.

A crucial aspect of this model is that the weight of each type of cue depends on context — more precisely, on its perceived reliability. When sensorimotor cues are clear and consistent (e.g., precise and real-time visual feedback), they receive a higher weight. When these cues are weak, ambiguous, or inconsistent, the weight shifts toward contextual and cognitive cues (e.g., expectations, beliefs, social context). This flexibility explains why SoA can be maintained even under conditions in which one type of cue is disrupted (Moore & Haggard, 2010).

### **3.4 Predictive Processing**

The fourth theoretical model — predictive processing — extends Bayesian integration to a hierarchical architecture of the brain. According to this framework, the brain is a system that continuously generates predictions about the hidden causes of sensory input and attempts to minimise prediction error — the discrepancy between prediction and actual feedback (Apps & Tsakiris, 2014; Friston, 2010). SoA, in this view, is not a separate module, but the result of integrating prediction errors at multiple hierarchical levels, from low-level sensorimotor levels to high-level cognitive levels.

A central concept in predictive processing is precision weighting. Not all prediction errors are treated equally: the brain assigns each source of information a precision (the inverse of its perceived variability), which determines how much that error matters in updating predictions. When a sensory signal is clear and consistent, its precision is high, and the associated prediction error carries greater weight. When the signal is ambiguous or inconsistent, its precision is low, and the error carries less weight (Apps & Tsakiris, 2014; Friston, 2010).

## **4. Components and Classifications of SoA**

### **4.1 The Feeling of Agency (FoA) and the Judgment of Agency (JoA)**

An important distinction in the conceptual delimitation of SoA is that between the feeling of agency (FoA) and the judgment of agency (JoA) (Gallagher, 2012; Synofzik, Vosgerau & Newen, 2008b; Bayne & Pacherie, 2007). According to this dual model, SoA is not a unitary phenomenon, but manifests at two distinct levels of processing.

FoA is an experience represented by an immediate, pre-reflective, and non-conceptual sensation, which does not require conscious reflection or deliberation. It arises during action preparation, from the sensorimotor prediction of effects, and is closely tied to immediate bodily signals (proprioception, efference copy). In other words, FoA is the raw sensation that "I am doing this" — a sensation that is not directly accessible to introspection and that manifests through behaviour, not through reflection. It answers the question: "do I feel that I am doing this?"

JoA, by contrast, is a high-level process — a reflective, conceptual, and controlled judgment that arises after the outcome. It relies on an interpretation about the authorship of a particular action, constructed from comparing prediction with actual feedback and integrating contextual information (Synofzik et al., 2008a). JoA involves concepts (causality, authorship, intention) and beliefs about the link between action and outcome. It answers the question: "do I believe that I did this?"

It is important to emphasise that the two levels occupy different positions in the hierarchy of cognitive processing, without interpreting this in terms of importance or quality of experience. FoA is defined mainly by sensory input and bodily signals, whereas JoA is closer to complex cognition, concepts, and inferences. Both levels are part of SoA, but they operate differently and can be influenced by different factors. The two aspects are not always aligned and can be empirically dissociated: in studies on passive movements, participants did not report FoA for movements executed by a mechanical device, but they did report JoA — the belief that the action could have been theirs — based on the experimental context (Dubynin & Shishkin, 2017).

## **4.2 Prospective and Retrospective Cues: Mapping onto the Experimental Design**

A third classification of SoA concerns the temporal moment at which a certain type of information contributes to agency. We can distinguish between prospective cues, which appear before or during the action, and retrospective cues, which appear after the outcome. This distinction is directly relevant to the design of the present study, because it maps onto the four components of SoA that the experimental manipulation was designed to isolate.

Prospective cues are signals available before or during the action, which anticipate the outcome and can determine SoA even before the effect is observed. These include action fluency (how easily and naturally response selection and preparation unfold), intention clarity (how precisely the action goal is mentally represented), and motor preparation (the readiness state of the motor system). Chambon and Haggard (2012) demonstrated that action fluency predicts SoA

independently of actual motor performance: when response selection is fluent, participants report stronger SoA, even if the action outcome is identical. In a subsequent study, Chambon, Moore and Haggard (2015) showed that transcranial magnetic stimulation (TMS) over the supplementary motor area (SMA) disrupts intentional binding, suggesting that the SMA is involved in generating the prospective predictions that support SoA. In the present design, the prospective component is captured by initiation time — the latency between face onset and the participant's action — which reflects motor preparation, response selection fluency, and anticipation of the effect's timing.

Retrospective cues, by contrast, are signals available after the outcome, which contribute to SoA through outcome monitoring and action-effect matching. These include comparing the initial prediction with actual sensory feedback, integrating contextual information about causality, and evaluating consequences. In comparator model terminology, retrospective cues correspond to detecting the match or mismatch between the efference copy and actual feedback (Frith, Blakemore & Wolpert, 2000). In Wegner's (2002) terminology, they correspond to the consistency criterion — the extent to which the observed outcome matches the initial intention. In the present design, the retrospective component is captured by the AOC condition (consistent/inconsistent/control), which reflects the match or mismatch between the intended direction and the observed direction of the avatar's gaze. This component corresponds to Wegner's (2002) consistency criterion.

A robust SoA does not rely exclusively on one type of cue, but integrates both sources of information, with their relative weight depending on context (Moore & Haggard, 2010). When prospective cues are clear and fluent, they can support SoA even before the outcome appears. When prospective cues are weak or ambiguous, SoA becomes more dependent on retrospective cues — on the match between what was intended and what was observed. This complementarity explains why SoA can be maintained even under conditions in which one of the two types of cues is disrupted.

Beyond these two temporal components, two further components of SoA must be considered in the experimental design. The first is the sense of ownership (SoO) — the awareness that one's own body and its parts belong to oneself (Braun et al., 2018; Tsakiris et al., 2007). In the present design, SoO is present in all conditions, including the control condition, because the participant always executes a motor response (pressing a key, moving the mouse, or touching the screen). For this reason, SoO cannot explain differences between conditions: it is constant and does not vary with the AOC manipulation. The second is intentionality / volition — the mental representation of the action goal. In the present design, this component is directly manipulated by AOC: in the consistent condition, the intention is fulfilled; in the inconsistent condition, the

intention is violated; in the control condition, the intention is absent. It is closely related to the retrospective component, as both are captured by the AOC condition.

This decomposition has two important methodological implications for data analysis. First, ANOVA on the AOC condition captures the full effect of agency — both the intentional component (action-outcome matching) and the temporal prediction component. Second, ANCOVA with initiation time as a covariate removes the temporal prediction component, thereby isolating the effect of the retrospective component. This distinction is crucial for interpreting the results: if the AOC effect remains significant after controlling for initiation time, then the effect can be confidently attributed to the retrospective component of SoA, rather than to temporal prediction.

An important methodological issue concerns SoO and the control condition. One might attempt to isolate SoO by comparing the action conditions (consistent, inconsistent) with the control condition. However, this comparison does not work: even in the control condition, the participant executes a motor action (pressing a key to identify the target), so SoO is not zero. The action vs. control comparison conflates SoO with intentionality and temporal prediction. The only clean comparison for SoO is across experiments — Exp 1 (keypress, minimal motor involvement) < Exp 2 (mouse, moderate involvement) < Exp 3 (pointing, maximal involvement). This methodological limitation is discussed in detail in the limitations section and in the general discussion.

## **5. Measurement of SoA**

### **5.1 Explicit Measures**

SoA can be measured in two distinct ways (Reddy, 2022). In explicit measures, participants verbally report whether they consider they have control over their actions or whether they (or someone else) are the causal agent of observed actions (Synofzik, Vosgerau & Newen, 2008a). Although verbal reports contain rich information, they can be influenced by subjective criteria, biases, or decision-making disturbances. For example, a liberal criterion can lead to more "yes" responses compared to a conservative criterion, but the liberal criterion is not necessarily linked to a stronger SoA (Wen & Imamizu, 2022). For this reason, researchers have developed indirect (implicit) measures.

### **5.2 Implicit Measures and Intentional Binding**

In implicit measures, researchers study how voluntary actions modify perceptual processes — such as temporal, visual, tactile, or auditory perception — without asking the participant to explicitly reflect on their agency (Haggard, Clark & Kalogeras, 2002; Moore & Obhi, 2012). The most commonly used implicit measure is intentional binding (or temporal binding). It refers to the perceived compression of the temporal interval between a voluntary action and its

sensory effect: when a person performs a voluntary action that produces a sensory effect, the interval between the two events is perceived as shorter than the objective interval.

More precisely, two temporal shifts occur simultaneously: (1) the action is perceived as occurring later than in reality (action binding), and (2) the effect is perceived as occurring earlier than in reality (effect binding). The two events thus approach each other in subjective time. Action binding and effect binding reflect different aspects of the intentional binding effect and are useful for separating anomalies in action perception from those in sensory feedback perception. For example, patients with Parkinson's disease show abnormal action binding, while their effect binding is similar to that of healthy participants (Wen & Imamizu, 2022). Importantly, the intentional binding effect is not observed when the action is involuntary — for example, when it is triggered by brain stimulation (Wen & Imamizu, 2022). This suggests that binding requires a volitional component, not just a simple temporal contingency.

The classical method for measuring intentional binding uses the Libet clock: a circular dial on which a dot rotates continuously. The participant must report the position of the dot at the moment they executed the action and at the moment they perceived the effect (Libet et al., 1983; Haggard et al., 2002). By comparing the reported position with the objective position, the subjective shift is calculated. This measure is considered implicit because it does not ask the participant to explicitly reflect on their agency.

### **5.3 Dissociations Between Explicit and Implicit Measures**

Although both types of measures are considered valid, they are not interchangeable. Wen, Yamashita, and Asama (2016) experimentally demonstrated that implicit and explicit measures can be dissociated: in a study that manipulated action-effect delay (0–1000 ms) and arousal (via stimulus colour), longer delay significantly reduced explicit agency ratings, but increased the intentional binding effect. Moreover, arousal amplified binding, but did not influence explicit ratings. This double dissociation suggests that the two types of measures reflect different aspects of SoA.

A similar dissociation was reported by Grynszpan, Sahaï et al. (2019) in a joint action study. Participants jointly manipulated two connected haptic devices, believing that their partner was always another person, although sometimes the partner was replaced by a robot. The results showed a striking dissociation: explicit agency judgments were higher with the robot partner, while intentional binding occurred only with the human partner.

A recent critical review of the literature on agency in joint action (Le Besnerais et al., 2024) confirms that implicit and explicit measures of SoA can diverge systematically. The authors emphasize that intentional binding and explicit agency judgments reflect different levels of processing and recommend using both types of measures for a complete characterization of SoA.

## 5.4 Critiques of Implicit Measures

Nevertheless, the validity of intentional binding as a purely implicit measure has recently been contested (Reddy, 2022). It has been argued that "implicit" reports are not perceptual effects generated by SoA, but judgment effects: under conditions of perceptual uncertainty — inherent in Libet clock tasks — participants rely on heuristics and response biases that are independent of the agentic state (see also Joordens et al., 2004, for the disappearance of the effect when temporal localisation is performed via mouse click instead of verbal report). In addition, many other factors — such as causal relations between events, attention, or arousal — can affect temporal perception and influence the binding effect (Wen & Imamizu, 2022).

Further evidence that binding is not a pure indicator of SoA comes from the study by Wen et al. (2016), who showed that arousal amplifies intentional binding without influencing explicit agency ratings. Moreover, it has been demonstrated that the intentional binding effect can occur even in the absence of intention: in a virtual reality study, simply passively viewing a virtual hand pressing a button to trigger a tone produced a binding effect similar to that in the condition where the tone was triggered by the participant's actual movements (Wen & Imamizu, 2022).

This critique has a direct implication for the present work: in a rapid colour-discrimination paradigm, with multiple trials and short temporal intervals, measuring intentional binding would be impracticable and ambiguously interpretable. For this reason, the thesis uses explicit measures (Likert), which allow direct comparison between conditions at the level of trial sets and are less susceptible to the methodological artefacts of implicit measures. However, the thesis also uses reaction times (RT) as an implicit, low-level indicator of agency-related processing, capturing the feeling of agency (FoA) rather than the judgment of agency (JoA).

## 6. Action-Effect Binding: The Mechanism Linking Action to SoA

A central concept linking the AOC manipulation to SoA is action-effect binding — the tendency to mentally associate actions with their anticipated sensory effects. According to ideomotor theory, actions are not mentally represented solely by their movements, but by the sensory effects they produce (Hommel, Müsseler, Aschersleben & Prinz, 2001). When a person learns to press a button to produce a sound, the mental representation of the action already includes the anticipated sound. This anticipated representation of the effect is activated before the action is executed and guides response selection and preparation.

Kühn, Brass and Haggard (2011) experimentally demonstrated that this action-effect link is directly relevant to SoA. In a series of studies, they manipulated the consistency between the participant's action and the observed effect, measuring both explicit SoA and intentional binding. The results showed that when the observed effect matched the anticipated effect, participants reported stronger SoA; when the effect was inconsistent with the anticipation, SoA was reduced.

Moreover, this modulation was present even when participants had no explicit intention to produce a specific effect — suggesting that action-effect binding operates at an automatic, pre-reflective level.

Applied to the design of the present study, action-effect binding provides the cognitive mechanism that explains why the AOC manipulation works. In the consistent condition, the participant intends a direction, activates the anticipated representation of the effect, and the observed effect matches this anticipation — which confirms the action-effect link and supports SoA. In the inconsistent condition, the participant intends a direction, but the observed effect is opposite — which violates the anticipated link, generates a prediction error, and reduces SoA. In the control condition, there is no action from the participant, hence no anticipated representation of the effect — and SoA is absent or minimal.

Thus, action-effect binding is not merely a theoretical phenomenon, but the mechanism through which AOC manipulates SoA. It explains why the consistency between action and effect is crucial: because actions are mentally represented in terms of their effects, and the match between the anticipated and observed effect is the condition for SoA to arise.

## **7. Clinical Implications: Schizophrenia and Autism**

Disturbances of SoA have been documented in various clinical conditions. In schizophrenia, patients may attribute their own actions to external agents or, conversely, may feel that they caused events over which they had no control — a phenomenon known as delusions of control (Frith, 2014; for a review, see Braun et al., 2018).

Less studied, but equally relevant to the present work, is the possible link between SoA and autism spectrum disorder (ASD). Individuals with autism often show particularities in processing social signals and in integrating sensory information, which could also affect the sense of agency. Although the literature is still limited, a few studies have reported subtle differences in SoA in individuals with autism, especially regarding the integration of sensory and motor cues (Zalla & Sperduti, 2015). For example, individuals with autism may have difficulty discriminating between their own actions and those of others, especially when sensory feedback is ambiguous or delayed (Grynszpan, Nadel et al., 2012). These observations are particularly interesting for the present study, because they suggest that SoA and gaze orienting — both fundamental components of social cognition — might interact in a way that has implications both for understanding basic mechanisms and for characterizing clinical profiles. We will return to this interaction in the section dedicated to modulators of the gaze cueing effect, discussing at length also the issue of theory of mind.

## **8. Conclusion: SoA as a Multifaceted Construct**

In this section, we have examined the concept of sense of agency (SoA) and its components, providing a theoretical framework for understanding how agency can modulate gaze cueing. We began with the definition of SoA as the subjective experience of being the initiator and controller of one's actions (Haggard & Tsakiris, 2009) and distinguished SoA from the sense of ownership (SoO), showing that the two can be dissociated (Braun et al., 2018; Tsakiris et al., 2007).

We then reviewed the four major theoretical models of SoA — the comparator model (Frith, Blakemore & Wolpert, 2000), Wegner's retrospective theory (2002), Bayesian integration (Moore & Haggard, 2010), and predictive processing (Apps & Tsakiris, 2014) — and argued that they are complementary, together providing a comprehensive framework for understanding SoA.

We then examined the main classifications of SoA: the distinction between the feeling of agency (FoA) and the judgment of agency (JoA) (Synofzik et al., 2008b); the distinction between prospective and retrospective cues (Chambon & Haggard, 2012); and the distinction between the different components of SoA — SoO, temporal prediction, intentionality, and action-outcome matching. We showed that these components map directly onto the design of the present study: RTs capture the implicit component (FoA), Likert ratings capture the explicit component (JoA), initiation time captures the prospective component, and the AOC condition captures the retrospective component.

We then examined the two ways of measuring SoA — explicit and implicit — and argued that they are not interchangeable, but reflect different aspects of agency (Wen et al., 2016; Grynszpan et al., 2019; Le Besnerais et al., 2024). Finally, we introduced the concept of action-effect binding (Hommel et al., 2001; Kühn et al., 2011) as the cognitive mechanism explaining why the AOC manipulation works.

The conclusion of this section is that SoA is a multifaceted construct that can be experimentally manipulated through the consistency between action and effect. The AOC manipulation in the present work is conceptualised as a direct manipulation of this consistency, and its effects on gaze cueing will be tested in the three experiments. In what follows, we examine whether agency — and particularly the SoA generated by AOC — can influence perception and attention, and whether this effect extends to social stimuli such as gaze.

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